

AI-Based Research Paper Analysis Using ResearchMind

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Abstract

This sample research paper presents ResearchMind, an AI-assisted system designed to help students and researchers analyze academic papers. The system uses natural language processing to extract important information from research documents, generate summaries, identify key findings, and answer questions about the uploaded paper. A retrieval-based question answering component is used to locate relevant passages before generating an answer.

1. Introduction

Academic researchers often need to read many papers to understand a research area. Long technical documents can make it difficult to quickly identify objectives, methods, results, and limitations. ResearchMind aims to reduce this difficulty by providing an interactive interface for paper analysis.

2. Research Objective

The primary objective is to develop an AI agent that can analyze uploaded research papers and provide reliable, easy-to-understand answers to user questions based on the document content.

3. Methodology

The system follows five major steps: PDF text extraction, document preprocessing, text chunking, semantic retrieval, and answer generation. The uploaded document is divided into smaller passages. When a user asks a question, relevant passages are retrieved and supplied to a language model to generate a grounded response.

Component	Purpose
PDF Processor	Extracts text from uploaded research papers
Text Splitter	Divides long documents into manageable passages
Embedding Model	Converts passages into numerical representations
Vector Database	Stores and retrieves relevant passages
AI Language Model	Generates summaries and answers

4. Results

In a small internal evaluation using 20 sample questions, the system successfully retrieved relevant sections for most questions. Users reported that summaries were easier to understand than manually scanning the full document. Questions requiring exact numerical details performed better when the relevant passage was explicitly retrieved.

5. Limitations

The system may produce incomplete answers when a paper contains complex tables, scanned images, mathematical notation, or poorly extracted text. The quality of generated answers also depends on the quality of the uploaded document and the underlying language model.

6. Future Work

Future versions could support multiple-paper comparison, citation graph analysis, improved table and figure understanding, multilingual paper analysis, and automatic identification of research trends across a collection of papers.

7. Conclusion

ResearchMind demonstrates how an AI agent can assist with academic paper analysis and interactive question answering. By combining document retrieval with language generation, the system can help users understand research papers more efficiently while keeping answers connected to the uploaded document.

Sample Paper Q&A;

Q1. What is the main objective of the paper?

To develop an AI agent that analyzes research papers and answers user questions using information from the uploaded document.

Q2. What technology is used for question answering?

The system uses semantic retrieval to find relevant passages and an AI language model to generate answers.

Q3. What are the main limitations?

Complex tables, scanned images, mathematical notation, and poor text extraction can reduce answer quality.